

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

firm regardless of AI exposure. For workers aged 22-25, we find a 12 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially.

Sixth, the above facts are largely consistent across various alternative sample constructions. We find that our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022, around the time of rapid proliferation of generative AI tools.⁴ They also hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we caution that the facts we document may in part be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

Why might AI adversely affect exposed entry-level workers more than other age groups? One possibility is that, by nature of the model training process, AI replaces codified knowledge, the “book-learning” that forms the core of formal education. AI may be less capable of replacing

⁴OpenAI introduced ChatGPT in November 2022.

consistent with the case studies above that show employment changes are visible across a range of occupations. Results with firm-time fixed effects in Figure 9 further show that our findings are robust to firm- or industry-level shocks that impact general hiring trends. These results indicate that our findings are not specific to technology roles.

Remote Work Figures A6 and A7 show results for occupations amenable to remote work (telework) and those that are not, according to Dingel and Neiman (2020).²² We find that, for young workers, more exposed occupations have slower employment growth, both in teleworkable occupations and in non-teleworkable occupations. The results for non-teleworkable occupations in particular suggest that our findings are not driven by outsourcing or work-from-home disruptions, at least solely.²³

Longer Sample Figure A8 shows results when extending the balanced sample of firms to 2018. This reduces the sample size and makes the data somewhat noisier. Nonetheless, the trends remain largely ordered by exposure in the post-GPT era, whereas this is not the case before 2022. A concern is that for the Eloundou et al. (2024) measures the most exposed quintile had slower employment growth starting around 2020. This is not the case for the Anthropic exposure measures, shown in Figures A9, A10, and A11. For these measures the most exposed groups have comparable employment growth throughout the period before generative AI, with divergent trends afterwards.

Changes in Education Another possibility is that the changes we observe are influenced by worsening education outcomes during the COVID-19 Pandemic. COVID-19 led to persistent harms in education outcomes (Kuhfeld and Lewis, 2025). As more educated workers on average have greater measured AI exposure, deterioration in the quality of education in recent years could influence the trends we observe.²⁴ In Figure A12 we show trends for occupations in which greater than

²²Whether an occupation is amenable to remote work is positively correlated with AI exposure. Only two teleworkable occupations fall in the lowest quintile of estimated AI exposure according to the GPT-4 β measure. Likewise few non-teleworkable occupations fall in the highest quintile of AI exposure. For this reason in Figure A6 we pool together the two lowest AI exposure quintiles into one group. In Figure A7 we pool together the two highest AI exposure quintiles.

²³Non-teleworkable occupations with high AI exposure include bank tellers, travel agents, and tax preparers.

²⁴Chandar (2025a) finds that declines in average skill levels for college graduates explain a sizable share of the slowdown in the growth of the college wage gap in recent decades.

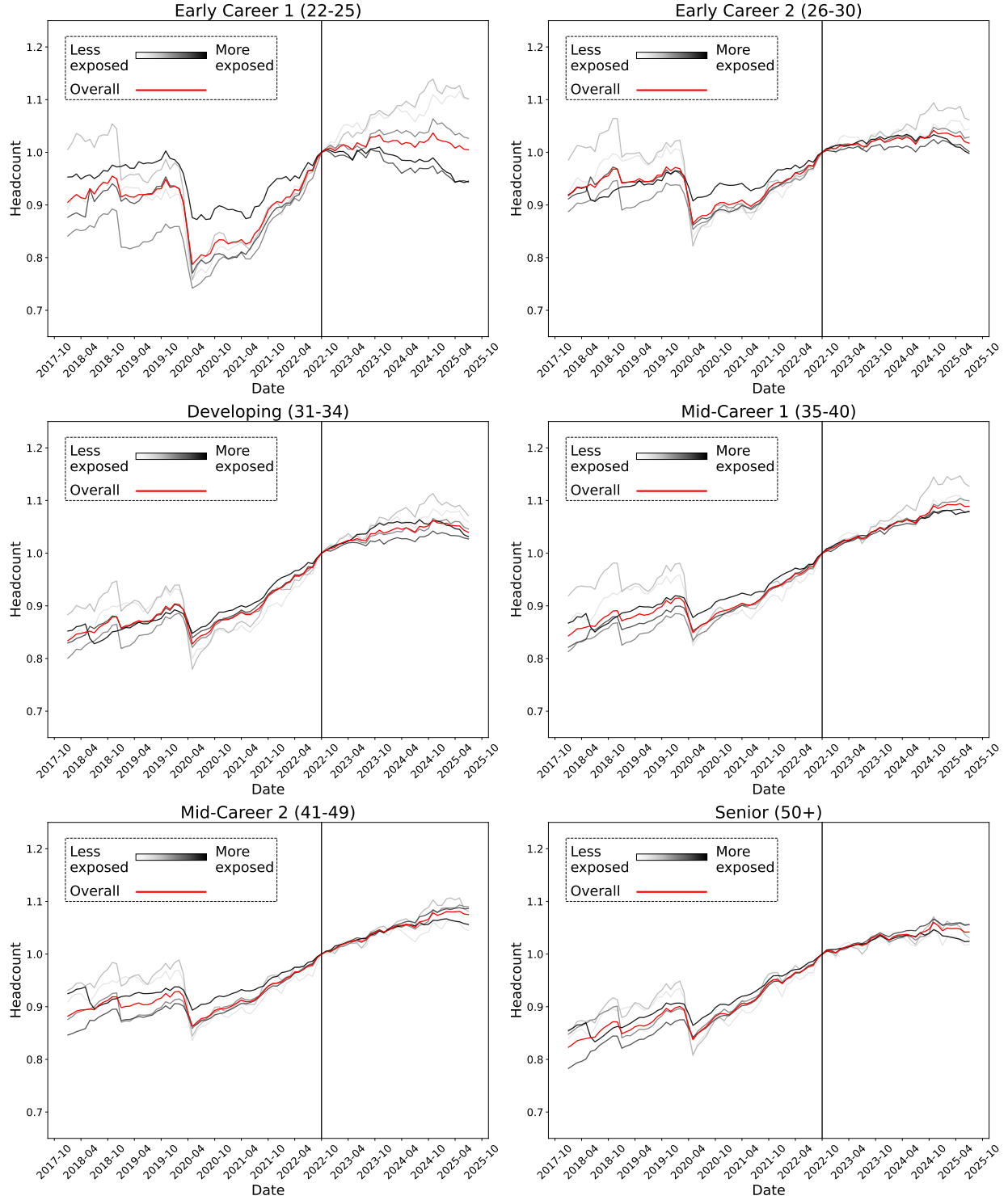


Figure A8: Employment changes by age and exposure quintile using measures from [Eloundou et al. \(2024\)](#). Data is from 2018 to 2025.

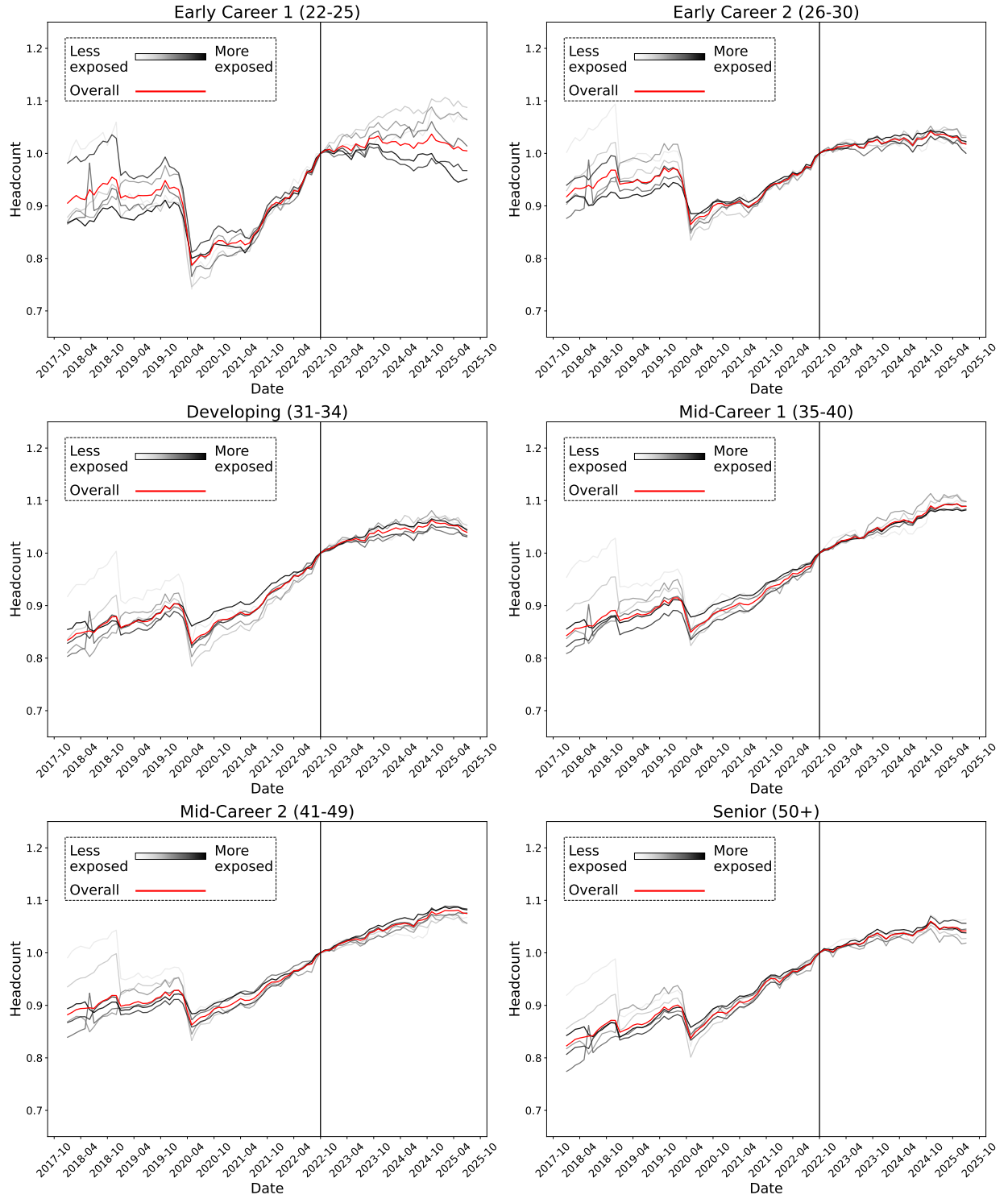


Figure A9: Employment changes by age and exposure quintile using Claude usage data from [Handa et al. \(2025\)](#). Occupations whose associated tasks all have fewer than the minimum number of queries to appear in the usage data are treated as a separate category, coded as 0. Data is from 2018 to 2025.

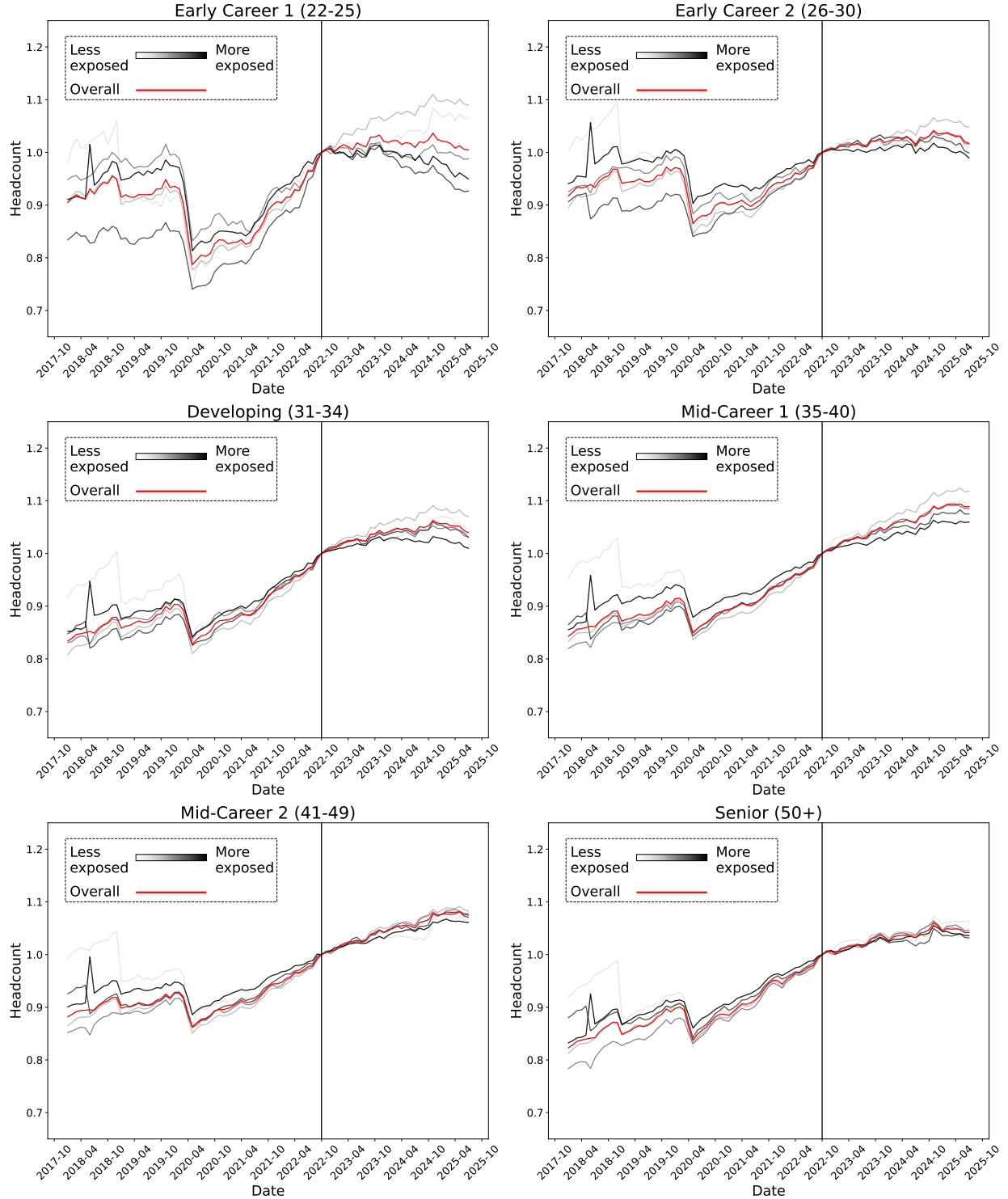


Figure A10: Employment changes by age and automation level using Claude usage data from [Handa et al. \(2025\)](#). Data is from 2018 to 2025.

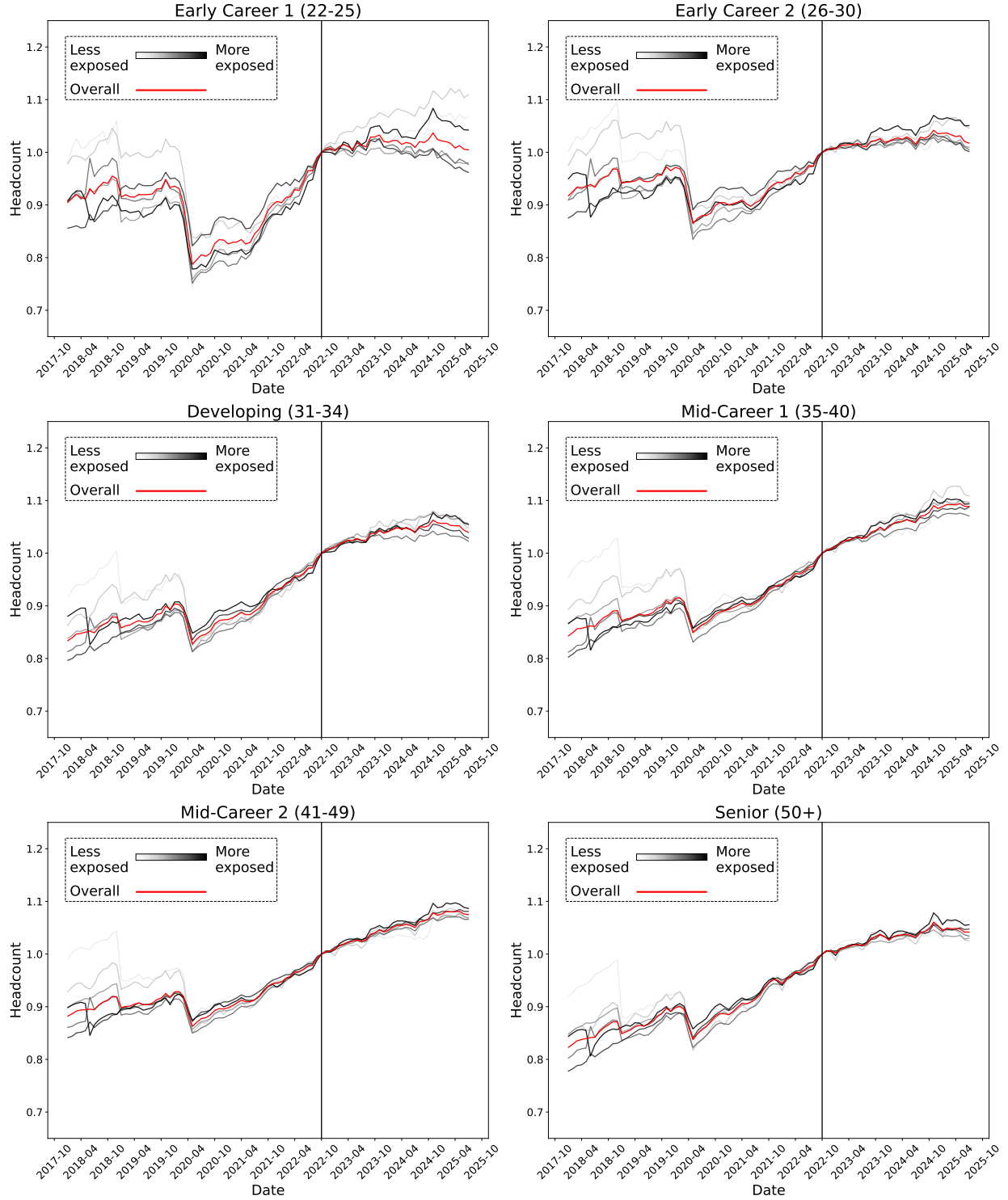


Figure A11: Employment changes by age and augmentation quintile using Claude usage data from [Handa et al. \(2025\)](#). Data is from 2018 to 2025.